How Age Affects the Second- and Third-Level Digital Divide: A Research Based on a Questionnaire-Based Survey

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Abstract

This questionnaire-based survey investigates the role of age in affecting the extent of the second- and third-level digital divide. Using an original questionnaire, the author collects 524 valid samples from all over China. With the extreme grouping method, total analysis, exploratory factor analysis (EFA), and confirmatory factor analysis (CFA), the validity of the items in the questionnaire is confirmed. Then, regression analysis and the intermediary effect analysis conclude that age is positively correlated with the extent of the second- and third-level digital divide. After that, the intermediary effect analysis identifies three intermediary variables: learning ability, interest in digital devices, and how sufficient people think their digital competence is.

Keywords: digital divide, age, questionnaire-based survey, regression analysis, intermediary effect analysis.

1. Introduction

The digital divide is now widely accepted as the gaps between differences in Internet access, Internet skills, and Internet outcomes. A great number of studies have been conducted to define the concept of the digital divide, to find the determinants of the digital divide, and to standardize terminologies.

Even with so many studies being done on the digital divide, there are still some limitations in past research. Firstly, although the digital divide is divided into three levels, the third level, the differences in Internet outcomes, is rarely studied. Secondly, most research doesn't focus on a single determinant of the digital divide, failing to explain the mechanism behind it. Lastly, most research is done in developed countries including the United States and the Netherlands, but the validity of the theories in developing countries has not been verified. Therefore, research focusing on how a single factor affects the second- and third-level digital divide in a developing country is essential to conduct.

Therefore, to bring forth new ideas based on past research, this article proposes and answers the following question: does age affect the second- and third-level digital divide, and how does age affect the second- and third-level digital divide? To answer this question, the author strives to (1) examine how age is correlated with the extent of the second- and third-level digital divide, (2) study how some variables are functioning as intermediary variables between age and the extent of the second- and third-level digital divide.

The article starts with a literature review, which introduces past research focusing on the concepts and determinants of the digital divide, followed by an explanation of the methodology being applied, and the result obtained. Finally, the conclusion and limitations of the study will be discussed.

2. Literature Review

2.1. The Definition of the Digital Divide: From Internet Access to Skills to Outcomes.

The digital divide became a topic of interest in 1995 when the National Telecommunications & Information Administration published the report "Falling Through the Net: A Survey of the Have Nots in Rural and Urban America". It was first defined as "the separation between those who have access to digital information and communications

technology (ICT) and those who do not (Dewan & Riggins, 2005)". Based on this definition, Dewan & Riggins considered the digital divide in three different levels, which are individual, organizational, and global.

As people more heavily depended on ICT and with strong support from the government, the Internet has developed rapidly since the 1990s and the share of the Internet population in the United States soon exceeded 50% by the beginning of the twenty-first century. As a result, some scholars pointed out that the distinction between those who have access to the Internet and those who don't was no longer the only manifestation of the digital divide, and that the differences in people's online skills should be taken into account (Hargittai, 2002). This was labeled the second-level digital divide while to make it easier to identify, the original definition of the digital divide was named the first-level digital divide. Internet use was later added to the concept of the second-level digital divide, which included the frequency and variety of Internet use (Scheerder et al., 2017).

With the deepening of the research, several researchers argued that in terms of the digital divide, the difference in the content which people create, also referred to as Internet outcomes, should be taken into account (Fuchs, 2009; Selwyn, 2004). This perspective of the digital divide was later named the third-level digital divide (Wei et al., 2011).

As the digital divide became no longer a binary but a comprehensive problem, researchers suggested that it is more appropriate to use the word "digital inequality" instead of "digital divide" (DiMaggio & Hargittai, 2001; Wei, 2012).

2.2. Determinants of the Digital Divide

Since the digital divide was first defined as the divide between haves and have-nots in Internet access, early research discussed various determinants based on how they affect Internet access. In a comprehensive analysis, the author summed up past research on the determinants of the digital divide in three levels including individual, organizational, and global. At the individual level, the most significant factors include income, location, race, age, and education (Hoffman, 1998). When it comes to the organizational level, the article suggests that, in past research, external pressure, organizational readiness, perceived benefits, federal grant, and demographic ethnicity are responsible for the first-level digital divide (Iacovou et al., 1995; Azari & Pick, 2005). At the global level, the determinants of the first-level digital divide include

human capacity, trading with OCED, property rights, the manufacturing share of the economy, infrastructure, wealth, and telecommunication policy (Caselli & Coleman, 2001; Quibria et al., 2003; Hargittai, 1999).

After the concept of the second- and third-level digital divide was proposed, other researchers did a systematic review of the determinants of the second and third-level digital divide. It is worth noticing that the second-level digital divide was divided into Internet skills and Internet uses, while the third-level digital divide remained to be Internet outcomes. The article examined the English-language academic digital divide literature between 2011 and 2016. The author divided the determinants of the digital divide into seven categories: sociodemographic, economic, social, cultural, personal, material, and motivational. As a result, the article suggested that the third-level digital divide has not been thoroughly studied, and the most commonly studied types of determinants that contribute to Internet skills and uses are sociodemographic and economic, while motivational factors are the most studied type of determinants in terms of Internet outcomes (Scheerder et al., 2017).

3. Methodology

3.1. The Design of the Questionnaire

The questionnaire includes three types of questions: completion, multiple choice, and scale. The questionnaire consists of 28 questions, including 11 6-point Likert scale items.

Question 1 asks the subjects to write down the year in which they were born, providing information about the independent variable, age. Questions 2-7 explore the control variables of the subject, which are gender, level of education, whether at work, and income. For question 4 ("What is your working state"), there are three choices, including "still a student", "at work", and "retired". If the subject chooses the first choice, the questionnaire will redirect the subject to question 5, asking about the average monthly income of their parents. If the subject chooses the second choice, the questionnaire will redirect the subject to question 6, which writes "What is your monthly income?" Finally, if the subject chooses the last choice, the questionnaire will redirect the subject to question 7, which is "What is your monthly pension". Question 8 ("Please choose 3 for this question") is an attention screening question. If the subject doesn't follow the question's instructions and chooses a number other than 3, the questionnaire will be voided. Questions 9-10 ask the subject to what extent they trust and are willing to use electronic devices.

To include as many contributors to the digital divide as possible, the questionnaire borrowed the categorization of the determinants of the digital divide from past researchers (Scheerder et al., 2017). In the systematic review, the authors summed up the most commonly-agreed determinants of the second- and third-level digital divide and put them into five categories, which are sociodemographic, economic, social, cultural, personal, material, and motivational. Among the five categories, sociodemographic, economic, and motivational are the only three that account for more than 10% of all studies. To make it possible to answer questions that provide information in the sociodemographic category questions 1-3 were addressed. As for the economic category and the motivational category, questions 4-7 and questions 9-10 were proposed. With questions 1-10 (except question 5 and question 8) all regarding the independent and controlled variables, all of them have corresponding variable names. The detailed questions 1-10 (except question 5 and question 8) and their variable names are shown in Table 1.

Questions 11-21 are about the extent of the second- and third-level digital divide, presented with 6-point Likert scales. Adopted from Scheerder (Scheerder et al., 2017), the author separated the second-level digital divide into the differences in Internet skills and use, and left the third-level digital divide to be the gap in Internet outcomes. The detailed items are shown in Table 2.

Finally, questions 22-28 strive to collect data regarding the possible intermediary variables. After open interviews with people from different age groups, the author came up with six possible-intermediary variables including learning ability, degree of myopia, degree of presbyopia, interest in digital devices, how sufficient they think their Internet competence is, and how many years they have used their mobile phones.

Table 1

Questions 1-10 (except question 4 and question 8) and their corresponding variable names

Question No.	Question	Variable name
1	In which year were you born?	Age
2	What is your gender?	Gender
3	What is your education level?	Education
5	What is your parents' average monthly income?	Parents' income
6	What is your average monthly income?	Average income

7	What is your average monthly pension?	Pension
9	Do much do you trust electronic devices?	Trust
10	How willing are you to use electronic devices?	Willingness

Table 2

Items and their item code

Items	Item code
How many functions do you know about social media such as WeChat and TikTok?	IS1
How good are you at distinguishing the authenticity of information?	IS2
What is the complexity of your website password?	IS3
How would you evaluate your Internet skills?	IS4
How would your family evaluate your Internet skills?	IS5
How long do you use your mobile phone every day?	IU1
How many kinds of applications do you open almost every day?	IU2
How frequently do you post WeChat moments, comments, and bullet screens?	IO1
How many likes do your WeChat moments, comments, and bullet screens get?	IO2
How frequently do you post videos and articles?	IO3
How many likes do your videos and articles get?	IO4

Note. IS stands for Internet skills, IU stands for Internet use, and IO stands for Internet outcomes. The factors will be confirmed in the CFA.

3.2. Extreme Grouping Method

The extreme grouping method is used to see whether an item can effectively distinguish between the group with the higher overall score and the group with the lower overall score or not. In other words, if the person with a higher score for a certain question is a lot more likely to get a better overall score, it means that this question is discriminative. The results can be seen in Table 3. With Sig. (2-tailed) of every item is less than 0.05, all the items are discriminative and would be kept in this standard.

Table 3

Results of the Extreme Grouping Method

Item code	Sig. (2-tailed)
IS1	.000
IS2	.000
IS3	.000
IS4	.000
IS5	.000
IU1	.000
IU2	.000
IO1	.000
IO2	.000
IO3	.000
IO4	.000

3.3. Total Analysis

Another method of examining the discrimination power is the total analysis approach. By relating the question's score with the total score, the total analysis approach presents the Pearson correlation coefficient which shows the discrimination power of the question. The results can be seen in Table 4. Usually, when the Pearson correlation coefficient with the total score of certain items is greater than 0.3, it means that the item is valid and will be kept. In this circumstance, all eleven items are valid and will be kept.

Table 4

Results of the Total Analysis

Item code	Pearson correlation coefficient with the total score
IS1	.687
IS2	.657
IS3	.646
IS4	.783
IS5	.743
IU1	.441
IU2	.537

IO1	.604
IO2	.532
IO3	.596
IO4	.571

3.4. Exploratory Factor Analysis (EFA)

After confirming that all the items were indeed discriminative, the author import the data into StataMP 17.0 to carry out the EFA. As a result, the p-value from the Bartlett test of sphericity is 0.000 and KMO equals 0.818. Normally, the data is suitable for carrying out the EFA when the p-value is less than 0.05 and KMO is greater than 0.6. Using paralleled analysis (PA), three factors were obtained, which had Eigenvalues > 1. The three factors were named as follows: Factor 1: Internet skills (IS1-5) with five items with a factor loading from 0.7331 to 0.8717. Factor 2: Internet uses (IU1-2) with two items with a factor loading from 0.7181 to 0.8187. Factor 3: Internet outcomes (IO1-4) with four items with a factor loading from 0.6241 to 0.8387.

Table 5

Factor Loadings of the Items from the questionnaire

Item code	Factor 1	Factor 2	Factor 3
IS1	.7522		
IS2	.7849		
IS3	.7331		
IS4	.8717		
IS5	.8198		
IU1			.8187
IU2			.7181
IO1		.6775	
IO2		.6241	
IO3		.8140	
IO4		.8387	

By figuring out the coefficient of each factor, and by adding up the product of the score for each factor and the coefficient of the corresponding coefficient, an overall score of one's digital competence regarding Internet skills, Internet use, and Internet outcomes can be calculated. The score for the three factors is named *f1*, *f2*, and *f3*, and *f* is used to denote the overall score. Altogether, four new variables were formed and would be added to further analysis.

3.5. Confirmatory Factor Analysis (CFA)

Based on the factors and the categorization of items obtained from EFA, CFA was carried out. The first step is to construct a structural equation model (SEM). However, if the SEM was constructed according to the results obtained from EFA, many indexes would have gone wrong. After some revision, IO1 and IO2 were dropped. The resulting SEM is shown in Fig. 1.

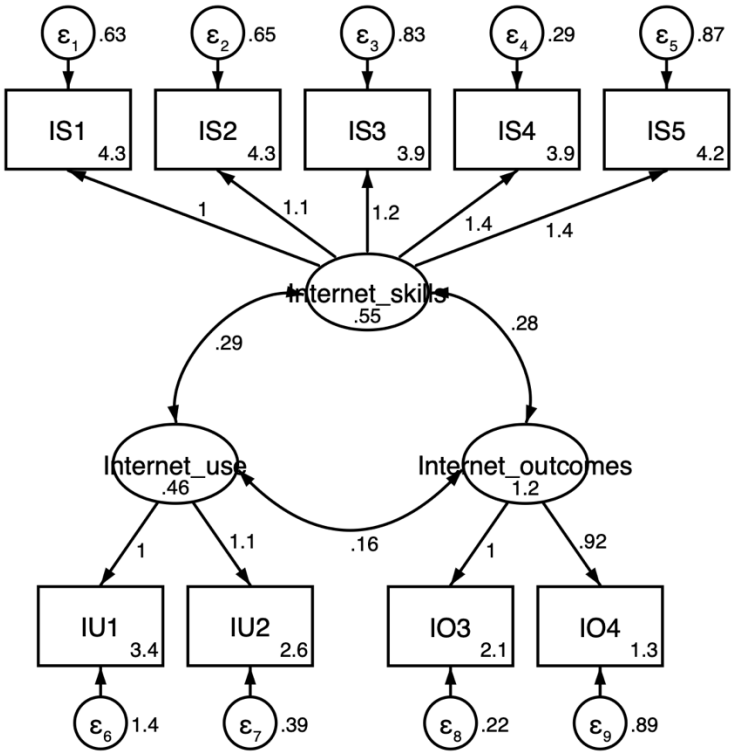

Fig. 1.

3.6. Regression Analysis

This article employs a linear regression analysis approach to find the answer to the research questions. The linear regression model can be represented as follows:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \epsilon$$

Where,

y denotes the overall score of one's digital competence regarding Internet skills, Internet use, and Internet outcomes, which is equilibrium to F found in EFA.
x_1 denotes the independent variable the research is focusing on, which is Age.
x_2 denotes one of the control variables, Gender.
x_3 denotes one of the control variables, Education.
x_4 denotes one of the control variables, Parents' income.
x_5 denotes one of the control variables, Average income.
x_6 denotes one of the control variables, Pension.
x_7 denotes one of the control variables, Trust.
x_8 denotes one of the control variables, Willingness.
$\beta_n (n \neq 0)$ denotes the coefficient, which can also be seen as the degree of contribution to the dependent variable, for the variable after it.
β_0 denotes the intercept of the function.
ϵ denotes the error term.

3.7. Intermediary Effect Analysis

To find out the intermediary variables between the extent of the second- and third-level digital divide and age, an intermediary effect analysis was carried out. The intermediary effect model can be represented as follows.

$$Y = cX + \varepsilon_1$$

$$M = bX + \varepsilon_2$$

$$Y = cX + dM + \varepsilon_3$$

Where,

Y denotes the overall score of one's digital competence, which is equilibrium to F found in EFA.

X denotes Age.

M denotes the intermediary variable.

a, b, c, and d denoted the coefficients.

$\varepsilon_n$denotes the error terms.

4. Results

4.1. Descriptive Analysis

4.1.1. Age

The questionnaire was collected from a variety of age groups: the minimum is 14 and the maximum is 87. There are 39 subjects aged 10-20 years old, accounting for approximately 7% of the total sample. There are 72 subjects aged 20-30 years old, accounting for approximately 14% of the total sample. There are 96 30–40 year-olds, accounting for approximately 18% of the total sample. There are 148 40-50 year-olds, accounting for approximately 28% of the total sample. There are 103 50-60 year-olds, accounting for approximately 20% of the total sample. There are 40 60-70 year-olds, accounting for approximately 8% of the total sample. There are 21 70-80 year-olds, accounting for approximately 4% of the total sample. There are 5 80-90 year-olds, accounting for approximately 1% of the total sample. The detailed distribution of age groups is shown in Fig. 2.

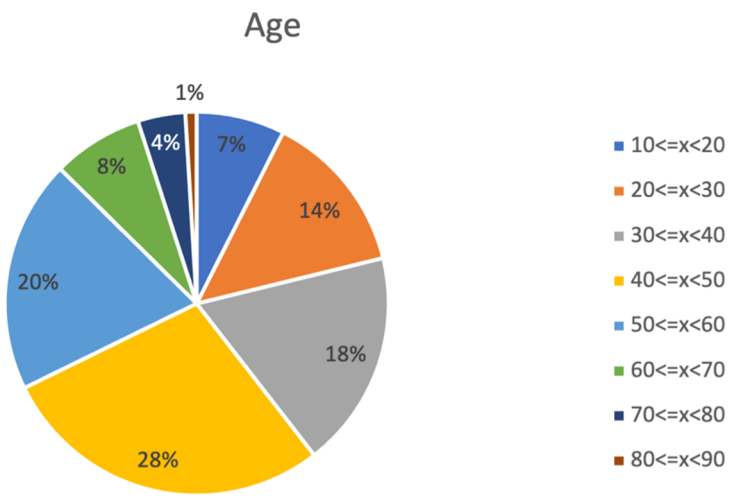

Fig. 2.

4.1.2. Gender

The male and female ratio is approximately equal. Of the 524 samples collected, there are 264 males, while the frequency of female subjects is 260. The detailed distribution of gender is shown in Fig. 3.

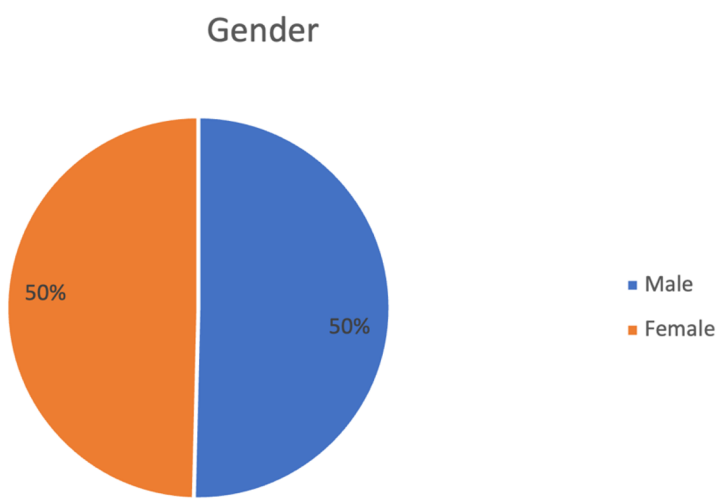

Fig. 3.

4.1.3. Education

Of the 524 samples collected, most people receive senior high school or a secondary specialized school education and university or junior college education, accounting for 83% of the total sample. 2 people did not receive any formal education, accounting for around 0% of the total sample. 3 people received primary education, accounting for 1% of the total sample. 18 people received junior high school education, accounting for 3% of the total sample. 290 people received senior high school or specialized school, accounting for 55% of the total sample. education, accounting for 55% of the total sample. 144 people received a university or junior college education, accounting for 28% of the total sample. 67 people received master's and above education, accounting for 13% of the total sample. The detailed distribution of education levels is shown in Fig. 4.

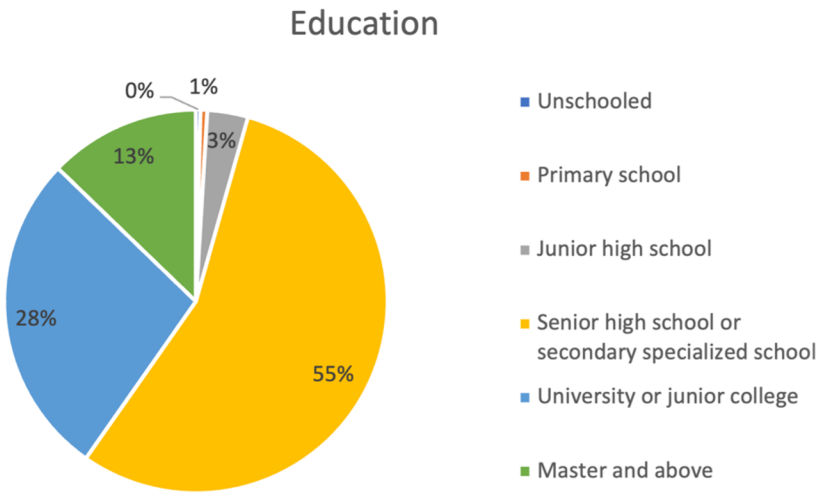

Fig. 4.

4.1.4. Working state

Of the 524 samples collected, 68 subjects reported being students, accounting for 13% of the total sample (see Fig. 5.). Of the 68 students, 15% reported their parents' average income is below 5,000 yuan, 19% reported their parents' average income is between 5,000 to 10,000 yuan, 13% reported their parents' average income is from 10,000 to 15,000 yuan, 10% reported their parents' average income is from 15,000 to 20,000 yuan, 10% reported their parents' average income is from 20,000 to 25,000 yuan, 12% reported their parents' average income is from

25,000 to 30,000 yuan, and the rest 21% reported their parents' average income is above 30,000 yuan (see Fig. 6.)

The majority of subjects reported that they were still working, and the proportion of people at work is 70%. Of the 368 people at work, 18% reported their monthly income is below 5,000 yuan, 37% reported their monthly income is from 5,000 to 10,000 yuan, 21% reported their monthly income is from 10,000 to 15,000 yuan, 9% reported their monthly income if from 15,000 to 20,000 yuan, 5% reported their monthly income is from 20,000 to 25,000 yuan, 2% reported their monthly income is from 25,000 to 30,000 yuan, the rest 8% their monthly income is 30,000 yuan and above (see Fig. 7.).

The rest of the 17% of the subjects reported being retired. Of the 88 retired subjects, 10% reported their monthly pension is below 2,000 yuan, 33% reported their monthly pension is from 2,000 to 4,000 yuan, 26% reported their monthly pension is from 4,000 to 6,000 yuan, 14% reported their monthly pension is from 6,000 to 8,000 yuan, 9% reported their monthly pension is from 8,000 to 10,000 yuan, and the rest 8% reported their monthly pension is 10,000 yuan and above (see Fig. 8.).

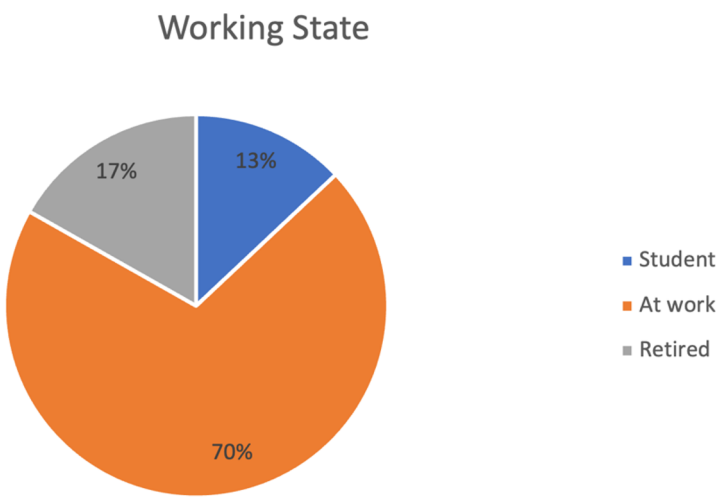

Fig. 5.

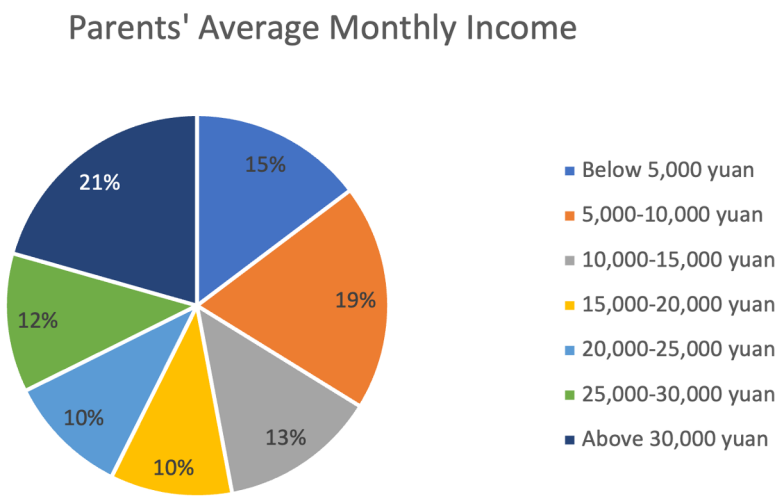
Parents' Average Monthly Income
15%
19%
13%
10%
10%
12%
21%
Below 5,000 yuan
5,000-10,000 yuan
10,000-15,000 yuan
15,000-20,000 yuan
20,000-25,000 yuan
25,000-30,000 yuan
Above 30,000 yuan

Fig. 6.

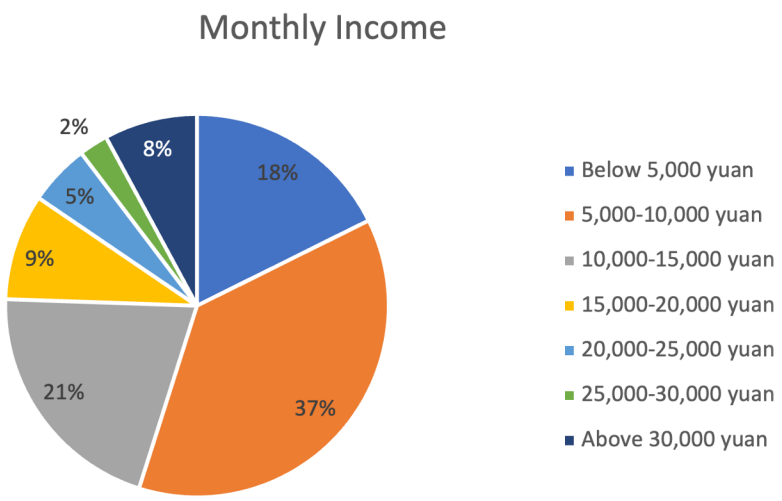
Monthly Income
18%
37%
21%
9%
5%
2%
8%
Below 5,000 yuan
5,000-10,000 yuan
10,000-15,000 yuan
15,000-20,000 yuan
20,000-25,000 yuan
25,000-30,000 yuan
Above 30,000 yuan

Fig. 7.

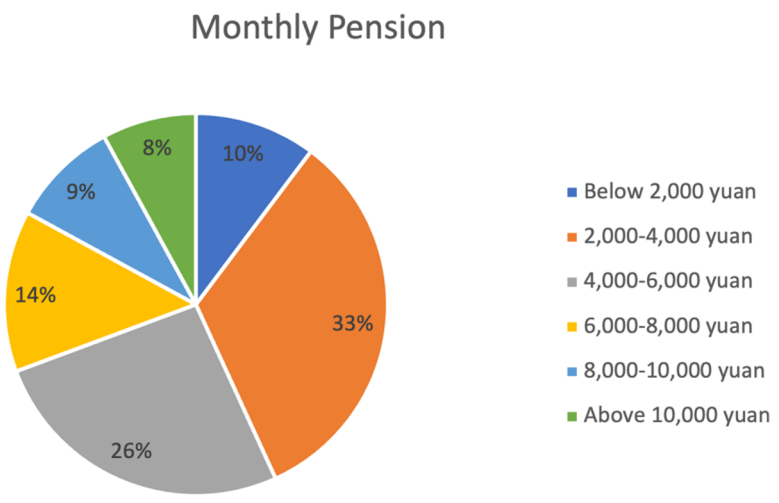

Fig. 8.

4.2. Results of Regression Analysis

4.2.1. Regression Analysis of the Overall Internet Competence

The overall Internet competence was calculated by the sum of 3 weighted factors, denoted by *f*. The results were shown in Table 6.

In the table, the P-value of Age, Parents' income, Trust and Willingness is less than .005, meaning that the effect of these variables on the dependent variable, *f*, is statistically significant. This indicates that age is indeed a predictor of overall digital competence. Since the coefficient of age is less than 0, it is concluded that as age goes up, digital competence goes down, which means the level of the digital divide goes up. Therefore, we can conclude that the extent of the digital divide is positively correlated with age. Other than that, average monthly income is also a strong predictor of overall digital competence. Lastly, the extent of the digital divide is negatively correlated with Trust and Willingness. In other words, people who trust digital devices more and are more willing to use them suffered less from the digital divide.

While some variables show a clear relationship with the extent of the digital divide, others cannot be regarded as predictors of the digital divide since their P-values far exceed .005. These kinds of variables include gender, education, parents' income, and pension. Although in a past systematic review, the author points out that gender was a determinant of the digital divide in past research (Scheerder et al., 2017), the status of men and women has become more equal

over the past couple of years, resulting in identical overall digital competence, which also means the identical extent of the digital divide. Education is also an often-mentioned determinant of the digital divide (Scheerder et al., 2017), but in Table 4 it doesn't seem like a strong predictor. As concluded previously, with age increasing, overall digital competence decreased. This can mean that the people with the highest overall digital competence are young people whose education levels are merely high school. While average income is a statistically significant predictor of overall digital competence, parents' income and pension are not, suggesting parents' average monthly income and pension are not proper variables for representing economic status and should be reconsidered.

Table 6

Results from Regression Analysis of the Overall Internet Competence

f	Coefficient	$P > \|t\|$
Age	-.0135	.000
Gender	.0291	.512
Edu	.0526	.056
Parent's income	.0054	.767
Average income	.0674	.000
Pension	.0476	.056
Trust	.0755	.002
Willingness	.1941	.000
_cons	-1.1250	.000

4.2.2. Regression Analysis of the Extent of the Digital Divide Regarding Internet Skills

The results are shown in Table 7.

The main difference between the results from regression analysis of the extent of the digital divide regarding Internet skills and results from regression analysis of overall digital competence is the difference regarding gender. While in Table 4 the P-value of gender is greater than 0.05, the P-value of gender in Table 7 is less than 0.05, indicating that males are better than females in terms of Internet skills (males are denoted with 1 while females are denoted with 0). Other results remain the same.

Table 7

Results from Regression Analysis of the Extent of the Digital Divide Regarding Internet Skills

f1	Coefficient	$P > \|t\|$
Age	-.0170	.000
Gender	.2535	.001
Edu	.0881	.054
Parents' income	.0413	.175
Average income	.1029	.000
Pension	.0980	.018
Trust	.1119	.006
Willingness	.3326	.000
_cons	-2.2252	.000

4.2.3. Regression Analysis of the Extent of the digital divide regarding Internet Use

The results are shown in Table 8.

While most results remain unchanged compared to the results from the regression analysis of overall digital competence, one changed particularly, and that is gender. From Table 8, it is easily observed that the coefficient of gender is a negative number, meaning males are likely to use digital devices for a shorter time and use fewer applications.

Table 8

Results from Regression Analysis of the Extent of the digital divide regarding Internet Use

f2	Coefficient	$P > \|t\|$
Age	-.0056	.000
Gender	-.0907	.001
Edu	.0367	.054
Parents' income	.0235	.175
Average income	.0323	.000
Pension	-.0047	.018
Trust	.0862	.006

Willingness	.0013	.000
_cons	-.2748	.000

4.2.4. Regression Analysis of the Extent of the Digital Divide Regarding Internet Outcomes

The results are shown in Table 9.

These results show that the people who are younger, wealthier, and more willing to use digital devices are on the advantaged side of the digital divide regarding Internet outcomes.

Table 9

Results from Regression Analysis of the Extent of the digital divide regarding Internet Outcomes

f3	Coefficient	$P > \|t\|$
Age	-.0181	.000
Gender	-.1532	.078
Edu	-.0044	.935
Parents' income	-.1081	.003
Average income	.0418	.140
Pension	.0151	.756
Trust	-.0271	.575
Willingness	.1854	.000
_cons	.0090	.982

4.2.5. Comparing the Results of Four Regression Analyses

From the four regression analyses, it is possible to tell that age is the most important predictor of the extent of the second- and third-level digital divide, whether from overall, Internet skills, Internet use, or Internet outcomes. Among the three perspectives of the digital divide, the extreme value of the coefficient of age regarding Internet skills and Internet outcomes all exceeds .01, while the coefficient of age regarding Internet use is only .0056. This indicates that the main gap between the youth and old is not about how long they use their phone or how many apps they open every day, it's about how good their Internet skills are and how much and in what quality they produce content with the Internet.

4.3. Results from the Analysis of the Intermediary Effect

To figure out the variables bridging the age and the extent of the second- and third-level digital divide, the intermediary effect analysis was carried out, in which six possible intermediary variables will be examined to see whether they are intermediary variables.

The first step is to do a regression analysis of age as the independent variable and a possible intermediary variable as the dependent variable. As a result, among the six possible intermediary variables (learning ability, degree, presbyopia degree, interest, sufficiency, year of use), the P-value for degree, presbyopia degree, and year of use are greater than .05, meaning that they're not significant and will be dropped. Further analysis of the intermediary effect will be carried out with the remaining possible intermediary variables. In addition, one more regression analysis was carried out, in which *f*, the index indicating overall digital competence, was treated as the dependent variable and the possible intermediary variable as the independent variable. The results are shown in Table 10., Table 11., and Table 12.

Table 10

Results from the Analysis of the Intermediary effect, in which the learning ability is the intermediary variable.

	f	Learning Ability
Learning Ability	0.251***	
	(15.19)	
age	-0.00885***	-0.0184***
	(-4.62)	(-3.65)
_cons	-1.552***	1.662***
	(-9.15)	(3.72)
N	524	524
R-sq	0.57	0.22
Adj. R-sq	0.56	0.21

Table 11

Results from the Analysis of the Intermediary effect, in which interest is the intermediary variable.

	f	Interest
Interest	0.245***	
	(13.21)	
age	-0.00778***	-0.0233***
	(-3.86)	(-4.97)
_cons	-1.526***	1.597***
	(-8.63)	(3.85)
N	524	524
R-sq	0.53	0.41
Adj. R-sq	0.52	0.20

Table 12

Results from the Analysis of the Intermediary effect, in which sufficiency is the intermediary variable.

	f	Sufficiency
Sufficiency	0.154***	
	(8.98)	
age	-0.0112***	-0.0146***
	(-5.26)	(-2.69)
_cons	-1.458***	2.100***
	(-7.64)	(4.36)
N	524	524
R-sq	0.46	0.20
Adj. R-sq	0.45	0.19

From the three tables, since all the coefficients for the three variables are marked with three asterisks, all three possible intermediary variables, learning ability, interest, and sufficiency are indeed intermediary variables. This means that they change when age changes and the overall

digital competence also changes when they do change. As a result, the causal relationship between age and the extent of the second- and third-level digital divide is explained.

5. Discussion

5.1. Conclusion

The research proposes the following research questions: (1) how age is correlated with the extent of the second- and third-level digital divide, and (2) how some variables are functioning as intermediary variables of age and the second- and third-level digital divide. The author first designs a questionnaire to find the answer to the questions and sends it out to various people. A total of 524 valid samples are collected. Then, the extreme grouping method and total analysis approach are applied to the data collected to confirm each item's discrimination power. All of the items pass both tests and are examined with EFA and CFA. With EFA and CFA, three factors are created and two items are dropped. Also, from the coefficient obtained from EFA, an index for overall digital competence, *f*, is developed.

With regression analysis, the article can answer the first question. It is concluded that: (1) the extent of the second- and third-level digital divide is positively correlated with age; (2) age mainly causes differences in Internet skills and Internet outcomes. Some other findings are also worth mentioning, for example, sociodemographic factors including age, economic factors including income, and motivational factors including the willingness to use digital devices and the trust in digital devices are all negatively correlated with the extent of the digital divide. Another finding is that women are more likely to use digital devices longer than men do and are likely to use more applications

Then, an analysis of the intermediary effect is conducted to see how age could affect the extent of the digital divide. The results show that the intermediary variables include learning ability, interest in using digital devices, and how sufficient people think their digital competence is.

5.2. Limitations

The study has some limitations. First, the external validity needs to be further studied. Although samples are collected from a fairly large amount of people, and the subjects include all genders, incomes, and education levels, with the fact that the questionnaires are sent by the

author and his parent through their social media accounts, the data collected might not be representative enough.

Second, the variable representing the overall digital competence, *f*, is calculated along with EFA, however, two items are deleted after CFA. Since the two items deleted are included in the calculation of *f*, and EFA is carried out earlier compared to CFA, small but inevitable imprecision would occur.

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